

Problem Statement ID	SIH26228
Problem Statement Title	Trustworthy Computer Vision Integrity Assurance for Data, Models and Inference Outputs in Multi-Contributor Pipelines
Theme	Blockchain & Cybersecurity
PS Category	Software
Team ID	179631
Team Name	Team Collectors

Offline integrity assurance for contributed data, models and inference records — measured, not asserted.

Evidence base: CVIAF worktree 2fda6eab (branch task3-real-backbone), reviewed main 6cddfe0e. Every figure in this deck names its file and denominator.

CVIAF — Computer Vision Integrity Assurance Framework

PROBLEM & INSIGHT

- Datasets, models and inference logs arrive from parties we do not control — each can be poisoned, swapped or edited.
- Today's checks cover one link each, none covers the chain, and none works when the network is gone.
- Classifier-era defences do not transfer: phantom boxes and cloaked objects have no classifier analogue.

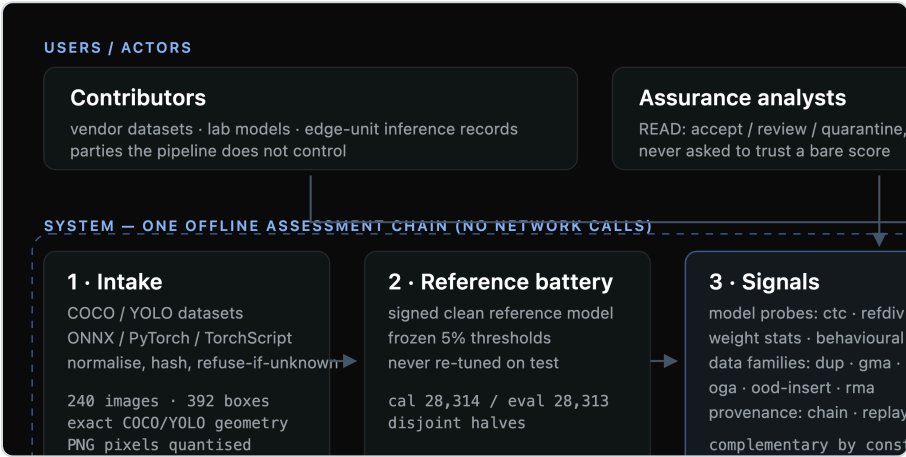
OUR SOLUTION

- One offline engine assesses data, model, inference records and drift, then merges every flag into one signed report.
- Ground truth first: a laboratory plants known attacks, so every detector is scored, not asserted.
- Every score is a conformal p-value against a signed reference battery we control; sample flags roll up per contributor.

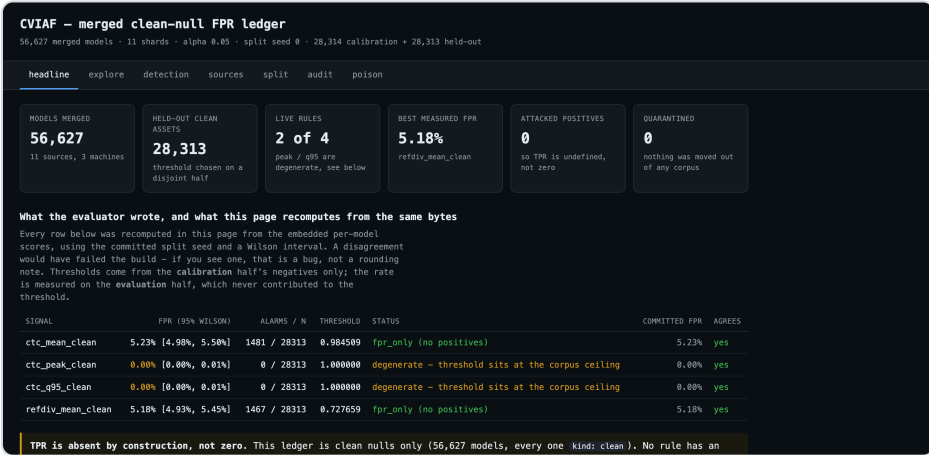
INNOVATION & PROOF

- Calibrated, not tuned — an explicit error guarantee, and accept is forbidden when an applicable check did not run.
- Cryptographic binding of image, model digest, config and output makes tamper, substitution and replay detectable.
- Imaged AUROC 0.9866 (fused_bonf, OGA fabrication) / 0.9062 (ODA cloaking); clean controls 0.500.
- Provenance: CVIAF 8/8 attacked records vs 4/8 for the enrolled-digest baseline; 0/2 genuine false alarms.
- One end-to-end report: 17 findings, CRITICAL → quarantine, 27.1 s audit-trail span (run oda_s5).

AUROC is an image-level ranking statistic on 8 laboratory models, not model-level recall. Provenance run signed with HMAC-SHA256 fallback, not Ed25519.



Architecture — users and actors over the offline chain. Everything inside the dashed trust boundary is computed from assets the team controls; contributors sit outside it.



Working measurement dashboard — committed clean-null ledger: 28,313 held-out negatives, alpha 0.05. Captured at committed settings; peak/q95 thresholds shown as degenerate.

TECHNICAL APPROACH

Every artefact entering a computer-vision pipeline — data, model, inference record — is measured against a reference we own. Detection is scored only at thresholds the clean ledger froze first: RefDiv-mean catches 47.9% (45/94) of substitution attacks at dose 0.25 and 93.7% (59/63) at dose 0.50, and 92.9% (92/99) of weight tamper at dose 1.00 — while CTC-mean, equally calibrated, catches almost none.

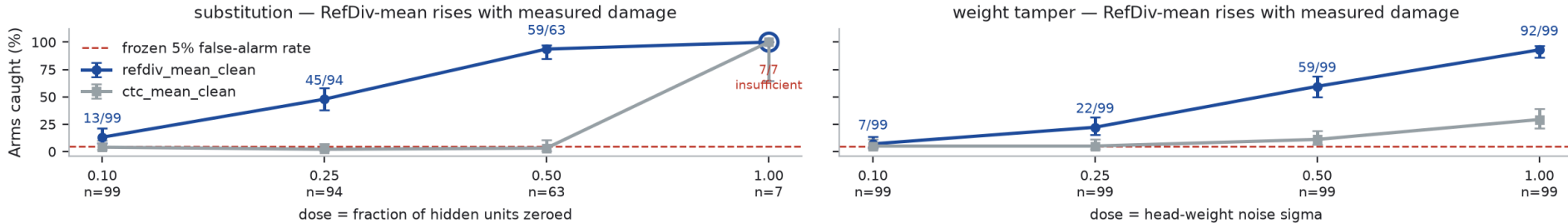
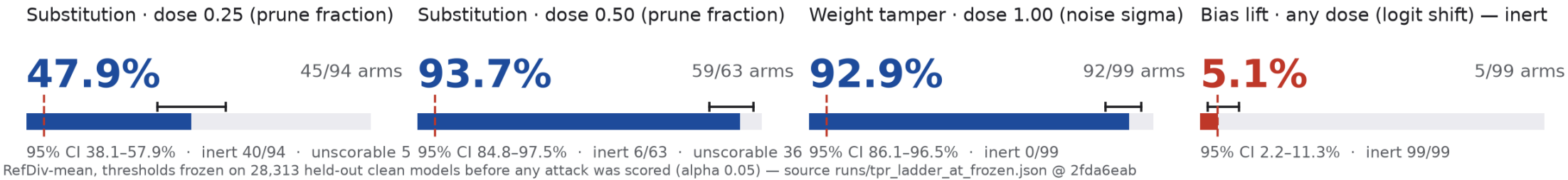
intake · reference

signals

frozen calibration

attack scoring

findings · coverage



Labels are caught/scored arms at the frozen threshold; hollow marker = below min_positives=20. Source: runs/tpr_ladder_at_frozen.json @ 2fda6eab

Finding	Evidence	Limit
Worked: RefDiv sensitivity rises with measured damage	substitution 13/99 → 45/94 → 59/63; weight_tamper 7/99 → 92/99. CTC-mean catches 2–29% at the same threshold.	thresholds frozen from the clean ledger; arms are synthetic TinyDetector models
Did not work: bias lift stays at the false-alarm rate	5/99 = 5.1% at every dose; 396/396 arms behaviour-inert (zero F1 change)	inert weight change, not a proven harmful backdoor — the blind spot is declared
Failure handled: unscorable arms are reported, not hidden	1,188 built → 1,055 scored; 133 unscorable, all substitution	substitution 1.00 dose has n=7, below min_positives=20 — shown hollow, never as 100%

Frozen thresholds, two populations kept apart — clean-null ledger: 28,313 held-out negatives; refdiv_mean_clean 5.18%, ctc_mean_clean 5.23%, both intervals including 5%; ctc_peak/q95 degenerate at the corpus ceiling. Coverage 14 of 14 clauses measured, gate_ok=True.

FEASIBILITY AND VIABILITY

Claim	Actual evidence	Boundary
Runs offline, end to end	verify-report validates runs/assurance/oda_s5 against the shipped schema; audit chain 7 entries, valid=True	laboratory corpus; no network in the assurance path
Ingests real dataset formats	COCO and YOLO round trip: 240 images, 392 boxes, exact annotation geometry	PNG pixels are quantised — not byte-equal
Produces a signed, auditable report	17 findings, CRITICAL → quarantine, 27.1 s; report schema-valid	HMAC-SHA256 fallback in this run — not Ed25519 non-repudiation
Named limits, not silent passes	14 supported attack classes and 8 unsupported conditions declared in the report's own coverage statement	demo run skips white-box checks and is kept separate (run demo_assurance)
Planned: real-backbone benchmark and deployment-grade signing	[needs value from H200 run ledger] — not measured in this repository	planned work, deliberately not presented as capability

```
zsh — CVIAF offline verification

$ python -m cviaf.lab verify-report \
    runs/assurance/oda_s5/assurance_report.json
valid against https://cviaf.local/schemas/assurance-report
# read from the same report (NOT a re-verification):
#   audit_trail_valid=True   length=7 entries
#   overall_risk=CRITICAL   disposition=QUARANTINE
#   audit-trail span 07:44:42 -> 07:45:09 = 27.1 s
```

```
zsh — CVIAF offline verification

$ python -m cviaf.lab coverage
# 14 of 14 clauses measured; gate_ok = True
# unscorable arms named, never silently dropped:
#   133 unscorable attack arms
# report declares 14 supported classes, 8 unsupported cond
```

Coverage resolved from live artefacts; unscorable arms named, not dropped.

Live verification of the named report (run oda_s5). This is a real command run on 2fda6eab.

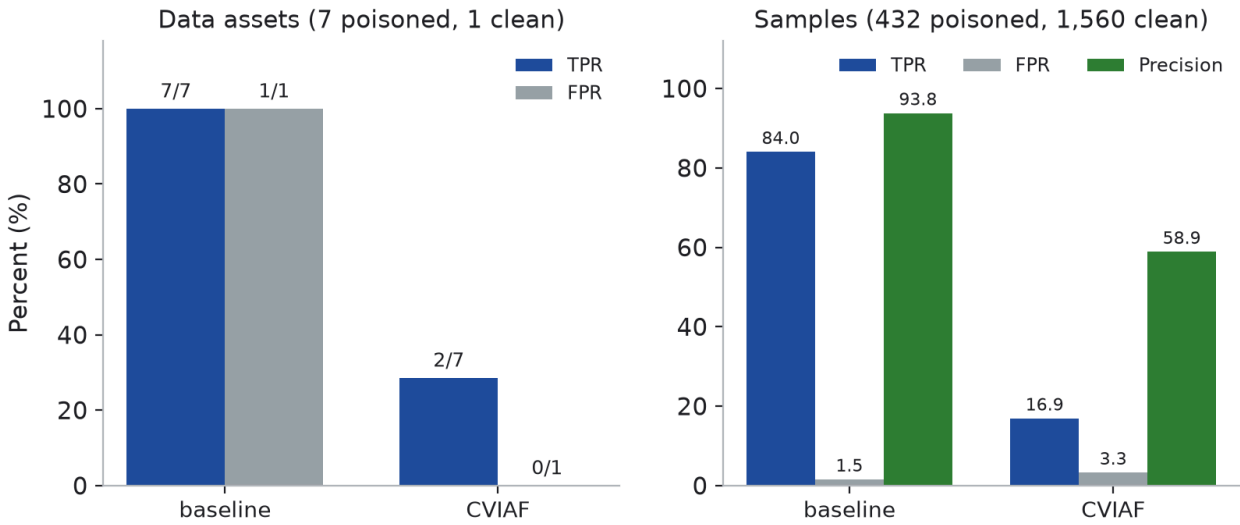
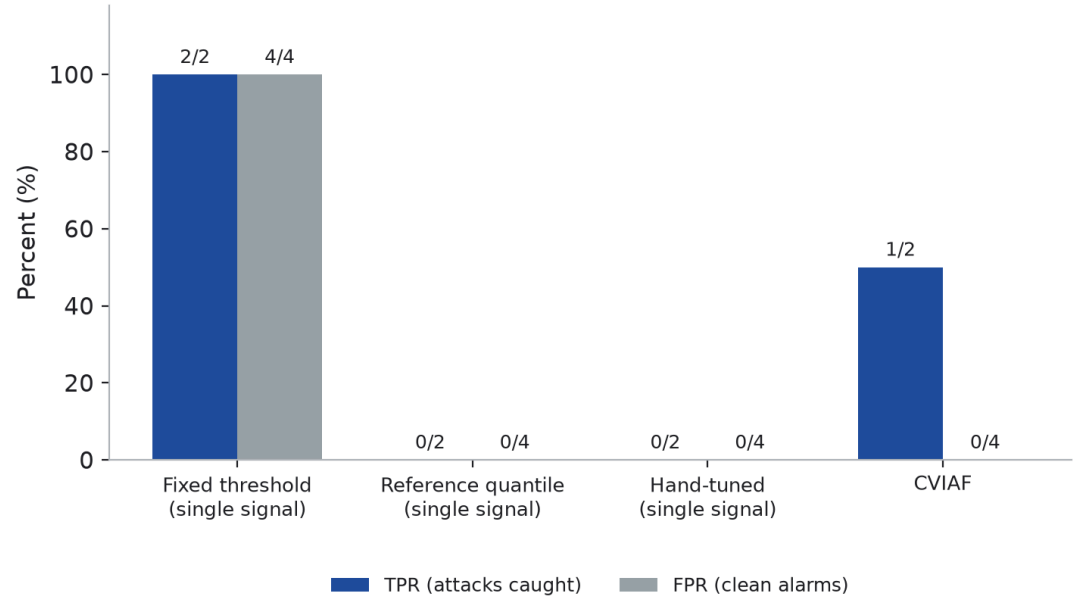
Export-inventory bar (22,693 members) is held in the supplement only: its 1,056 and 11,637 components are the seed_10150_28385 and seed_10264_24607 shards already inside the 56,627-model study, so the two populations are overlapping and must never be added. A dedup receipt is still outstanding.

[insert lab photo — AI Lab @ AU QUASAR; context only, not a benchmark]

IMPACT AND BENEFITS

Implemented baseline checks vs CVIAF on the same laboratory cases. This is a measured tradeoff on a small cohort, not a claim of universal superiority.

Same MVP cohort, model axis: 2 scored positives, 4 negatives, 2 ASR-gated exclusions



Source: runs/mvp/comparison.json @ 2fda6eab. CVIAF trades lower item recall for fewer false positives here — this is a tradeoff, not dominance.

Source: runs/mvp/comparison.json @ 2fda6eab. Tiny illustrative denominators — not a production guarantee.

What changed	Before (implemented baseline)	After (CVIAF)	Boundary
Model axis	fixed threshold 2/2 TPR, 4/4 FPR	1/2 TPR, 0/4 FPR	2 positives, 4 negatives, 2 ASR-gated excluded
Inference tamper + replay	enrolled digest 4/8; abstains 2 attacked, 2 genuine	8/8, FPR 0/2	HMAC fallback; tamper check, not proof of origin
Data asset triage	7/7 TPR but 1/1 false alarm	2/7 TPR, 0/1 false alarm	fewer asset false alarms, lower recall in this run
Sample-level decisions	TP 363 / FP 24, TPR 84.03%, precision 93.80%	TP 73 / FP 51, TPR 16.90%, precision 58.87%	unfavourable to CVIAF — retained in the supplement
Accountability	hash-only: top-1 undefined, 0/8 detected	top-1 2/2 on single-culprit classes	2 scored classes; not universal source accuracy

Read this as an engineering result: fewer asset false alarms and a richer evidence trail, paid for with lower item-level recall. The 56,627-model clean study measures false alarms only — it cannot produce TPR, precision or savings, and no measured reduction in analyst hours, incidents, cost or readiness risk exists in this repository.

RESEARCH AND REFERENCES

Item	Reference
PS 26228 problem statement	SIH 2026 official statement (attached deck, page 1)
Repository + commit	github.com/Collectors-AU/CVIAF @ 2fda6eab (branch task3-real-backbone)
Same-cohort comparison	runs/mvp/comparison.json — model, data and provenance axes
Large clean-null FPR receipt	runs/merged_fpr_tpr_report.json (56,627 models)
Fleet ledger	runs/fleet_fpr_ledger_report.json (7,503 models, 3,751 held-out)
Dose-ladder receipt	runs/tpr_ladder_at_frozen.json (1,188 built / 1,055 scored)
Coverage + as-built matrix	runs/coverage.json, docs/COVERAGE_STATEMENT.md, docs/PS26228_ALIGNMENT_MATRIX.md
Report example	runs/assurance/oda_s5/assurance_report.json (schema-valid, 17 findings)

Evidence manifest: evidence/evidence_manifest.json — every slide number maps to a file, field and denominator. All figures are laboratory/synthetic scope; field validation is pending. docs/PS26228_REQUIREMENT_TRACE.md is historical design intent, not as-built evidence.